Florian Pothin

Ph.D. candidate in Economics — climate and environmental economics, carbon accounting, trade and embodied carbon, heterogeneous-firm macroeconomics

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Education

Ph.D. in Economics, Université de Rennes (CREM) 2023 – 2026

Industrial (CIFRE) doctorate in partnership with Toovalu. Thesis: *The Decarbonization Gap: Firms, Nations, and the Economics of the Low-Carbon Transition*. Defence: 24 November 2026, Rennes. The thesis crosses the scales of climate action one at a time (firm, nation and trade, then the firms that abate) and asks at each step what the number measures before asking whether it moves. Recurring finding: disclosure, pledges, income growth and trade policy move the appearance or the location of emissions more than their quantity; a carbon price cuts them, and its revenue allocation decides how much. Each empirical paper has a Zenodo replication package.

- Supervisors: Fabien Rondeau and Tovonony Razafindrabe. Committee: Hamza Bennani (Nantes Université) and Marc Baudry (Université Paris Nanterre), rapporteurs; Aude Pommeret (Université Savoie Mont Blanc), president; Katheline Schubert (Université Paris 1 Panthéon-Sorbonne).

M.Sc. in Macroeconomics and Finance, Université de Rennes (graduated with honours) 2018 – 2020

B.Sc. in Economics, Université de La Réunion 2015 – 2018

Summer school "Tools for Macroeconomics", London School of Economics 2025

Research papers

Beyond Reported Emissions: Uncovering Hidden Corporate Carbon Footprints. *International Journal of Finance & Economics, under review, 2nd round*.

With F. Rondeau, T. Razafindrabe and A. Macherey. CDP panel 2013–2023, fifteen Scope-3 categories: detects which categories firms omit or under-report and estimates the missing magnitude from credible reporters. The hidden part is 12.6–22.0% of reported emissions (81.7 Gt), concentrated in the use of sold products.

Do Corporate Climate Pledges Cut Emissions? Science-Based Targets Change What Firms Count First. *working paper, revised for Ecological Economics; Research Square preprint*.

Staggered difference-in-differences on SBTi firms linked to CDP (2015–2023). In the year after a pledge a firm discloses 6.7 percentage points more of the Scope-3 categories material to its sector; reported Scope 3 rises 41% while emissions within the pre-pledge category boundary fall 32%; Scope 1 shows no lasting change. The sign of the measured effect depends on a boundary the firm chooses.

Trade Shocks Redraw the Global Carbon Map but Do Not Shrink It. *Review of World Economics, under review*.

Embodied carbon on a trade network under sanctions, tariffs and WTO accession (PPML, staggered-adoption estimators). Russian carbon exports to the sanctioning bloc fall 31% then 74% and reappear in third countries. The trade elasticity estimated on an actual tariff (1.1–2.1) is far below the calibrated 4–6; at that value a border carbon adjustment cuts covered-sector embodied carbon by 2–4%, and a third to all of the cut leaks.

Is the Environmental Kuznets Curve Identified? Turning Points under Production-, Consumption- and Trade-Adjusted Life-Cycle Emissions. *Energy Economics, submission in preparation*.

Strict identification criterion (bootstrap interval of the turning point inside the observed income support), size and power by Monte Carlo; parametric, semiparametric and neural estimators; replication audit of published neural-network turning points. No turning point identified for production- or consumption-based emissions; the life-cycle measure yields one, marginal and quadratic-only.

Endogenous Abatement as Green Capital in a Heterogeneous-Firm General Equilibrium. *with T. Razafindrabe; Journal of Environmental Economics and Management, in preparation*.

Hopenhayn economy in which abatement is the green share of two irreversible capital stocks and emissions attach to the fuel that runs the dirty one. A 42% cut runs mostly through selection (dirty firms shrink and exit); an auction cuts emissions 57% where a rebate indexed on own emissions cuts nothing; the welfare premium of free allocation is an entry-scale distortion, not a climate effect.

Awards and projects

Prix Transition, Trophées Valorisation 2024, Campus Innovation, Université de Rennes Nov. 2024

Awarded to T. Razafindrabe, F. Rondeau and F. Pothin for the CREM–Toovalu research project behind the thesis.

ANR I-DAVE, research contributor (Toovalu side) 2024 – 2027

AI and knowledge-based framework for the reliability of life-cycle assessment, with Centrale Nantes (LS2N), UTC (Roberval), Toovalu and Cetim: data harmonisation, emissions modelling, reliability assessment.

Presentations

- *International Economies: Challenges Ahead* (Ryukoku Univ. with Yokohama National Univ., Sciences Po Aix, AMSE), Kyoto, Feb.–Mar. 2026; research workshop, Shiga University, 2026 (trade shocks and embodied carbon).
- 11th International Macroeconomics Workshop, *International Macroeconomic Issues of Climate Change*, CREM, Rennes, 2025 (sectoral climate legislation).
- SESTEF 2024, Paris, Dec. 2024; FFM29, IPAG, Oxford, Sept. 2024 (hidden corporate carbon footprints).
- Practitioner talks: Produrable 2025 and 2026, Paris (footprint of AI); ADN Ouest and Boavizta, 2024 (spend-based emission factors).

Service and standards

- Member, Science Based Targets initiative Expert Working Group on Data Quality Assurance and Claims; member, IRIS-E (Université de Rennes), interdisciplinary research on the socio-ecological transition: evaluation of project proposals and funding decisions (from Dec. 2023).
- Pothin & Gilormini, *Multi-criteria environmental assessment of generative artificial intelligence services: a life-cycle-based method*, Zenodo, doi:10.5281/zenodo.22015676 (public methodological note).

Teaching

Macroeconometrics and Macrodynamics, M.Sc., Université de Rennes, 2023–2027 (≈120h). **Business and engineering schools**, 2022–2025 (≈150h): supervised machine learning in R (IA School), Python and AI frameworks (EPSI), mathematics for finance (IGEFI), green finance (IET).

Professional experience

Toovalu (Nantes), carbon-accounting software — Data scientist, Research & Impact Apr. 2022 – Aug. 2026

CIFRE doctoral position; independent consultant to the firm since Sept. 2026.

- Built the firm's emissions observatory (firm, sector and country level; CDP, ADEME, Bloomberg, EXIOBASE), whose sectoral-comparison module implements the correction framework of the first paper; production models for emissions estimation and Scope-3 gap filling in a B2B SaaS platform; internal lead on AI methods and on the footprint of AI.

Natixis Investment Managers (Paris) — Quantitative research intern Mar. 2020 – Jan. 2021

Return-prediction models for a multi-asset fund; ESG integration in portfolio construction. **CREM, Université de Rennes**, research assistant, Apr.–Aug. 2019: country-risk analyses for OPEC member states.

Skills and languages

Methods: heterogeneous-firm general equilibrium (value-function iteration, simulated-moments calibration); panel econometrics, staggered difference-in-differences, event studies, PPML, GMM/IV, bootstrap inference, power analysis. **Machine learning:** gradient boosting, random forests, regularised regression, neural and graph neural networks (scikit-learn, PyTorch, PyTorch Geometric). **Data:** CDP, SBTi, ADEME Base Empreinte, ecoinvent, EXIOBASE, BACI, Bloomberg, World Bank, OpenAlex; multi-source harmonisation, reproducible pipelines, Git, Zenodo. **Programming:** Python, R, MATLAB/Dynare, SQL, LaTeX. **Languages:** French (native), English (professional), German (intermediate), Japanese (A1).

References

Fabien Rondeau, Professor, Université de Rennes, CREM UMR CNRS 6211, fabien.rondeau@univ-rennes.fr | **Tovonony Razafindrabe**, Associate Professor (HDR), Université de Rennes, CREM UMR CNRS 6211, tovonony.razafindrabe@univ-rennes.fr